

Question 1

(a)

To show V is a subspace, we derive the homogeneous system from $AB = BA$. Given $A = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} w & x \\ y & z \end{pmatrix}$, compute AB and BA :

$$AB = \begin{pmatrix} w-y & x-z \\ w-y & x-z \end{pmatrix}, \quad BA = \begin{pmatrix} w+x & -w-x \\ y+z & -y-z \end{pmatrix}$$

Equating $AB = BA$ yields:

$$\begin{cases} x+y=0 \\ 2x+w-z=0 \\ w-2y-z=0 \end{cases}$$

The system reduces to two equations: $x+y=0$ and $w-2y-z=0$. Thus, V is a subspace with dimension 2.

(b)

Set $S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \\ 2 \end{pmatrix} \right\}$. Both vectors satisfy the homogeneous system. They are linearly independent and span V , hence form a basis.

(c)

Check each vector against the homogeneous system:

- $\mathbf{u}_1$: Substitute $w=1, x=-3, y=3, z=-5$:

$$\begin{cases} x+y = -3+3 = 0 & \checkmark \\ w-2y-z = 1-2(3)-(-5) = 1-6+5 = 0 & \checkmark \end{cases}$$

Coordinates: Solve $a \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ -1 \\ 2 \end{pmatrix} = \mathbf{u}_1$:

$$a=1, b=-3 \Rightarrow \text{Coordinates: } [1, -3].$$

- $\mathbf{u}_2$: Substitute $w=10, x=2, y=2, z=6$:

$$\begin{cases} x+y = 2+2 = 4 \neq 0 & \text{(Fails)} \end{cases}$$

Not in V .

- $\mathbf{u}_3$: Substitute $w = 7, x = 3, y = 3, z = 7$:

$$\begin{cases} x + y = 3 + 3 = 6 \neq 0 \end{cases} \quad (\text{Fails})$$

Not in V .

- $\mathbf{u}_4$: Substitute $w = 3, x = -2, y = 2, z = -1$:

$$\begin{cases} x + y = -2 + 2 = 0 & \checkmark \\ w - 2y - z = 3 - 2(2) - (-1) = 3 - 4 + 1 = 0 & \checkmark \end{cases}$$

Coordinates: Solve $a \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ -1 \\ 2 \end{pmatrix} = \mathbf{u}_4$:

$$a = 3, b = -2 \Rightarrow \text{Coordinates: } [3, -2].$$

- $\mathbf{u}_5$: Substitute $w = 2, x = 1, y = -1, z = 4$:

$$\begin{cases} x + y = 1 + (-1) = 0 & \checkmark \\ w - 2y - z = 2 - 2(-1) - 4 = 2 + 2 - 4 = 0 & \checkmark \end{cases}$$

Coordinates: Solve $a \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ -1 \\ 2 \end{pmatrix} = \mathbf{u}_5$:

$$a = 2, b = 1 \Rightarrow \text{Coordinates: } [2, 1].$$

$\mathbf{u}_1, \mathbf{u}_4, \mathbf{u}_5$ are in $\mathbf{V}$
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Question 2

(a)

Matrix with vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ as columns:

$$\begin{pmatrix} -1 & 3 & 0 & -2 \\ 5 & -1 & 5 & 3 \\ 3 & 1 & 4 & 1 \\ -2 & 0 & -1 & -1 \end{pmatrix}$$

Row reducing to RREF:

$$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There are 3 pivot columns, so the dimension of V is 3. The set is not a basis due to linear dependence.

(b)

From the RREF, the null space is spanned by $\begin{pmatrix} -1 \\ 1 \\ 0 \\ 2 \end{pmatrix}$. A valid 3×4 matrix A with linearly dependent rows orthogonal to this vector is:

$$A = \begin{pmatrix} -1 & 1 & 0 & 2 \\ 2 & -2 & 0 & -4 \\ 1 & -1 & 0 & 2 \end{pmatrix}$$

This ensures $A\mathbf{x} = \mathbf{0}$ defines V .

(c)

Vector $\mathbf{w} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ does not satisfy $-1(1) + 1(0) + 0(0) + 2(0) = -1 \neq 0$, hence not in V .

Question 3

(a)

$U \cup V$ is **not a subspace**. To be a subspace, $U \cup V$ must be closed under addition and scalar multiplication. Since neither $U \subseteq V$ nor $V \subseteq U$, there exist vectors $\mathbf{u} \in U$ and $\mathbf{v} \in V$ such that $\mathbf{u} + \mathbf{v} \notin U \cup V$. For example, adding the first vectors of U and V :

$$\begin{pmatrix} 5 \\ 5 \\ 4 \\ 7 \\ -2 \end{pmatrix} + \begin{pmatrix} 3 \\ 4 \\ 3 \\ 5 \\ -2 \end{pmatrix} = \begin{pmatrix} 8 \\ 9 \\ 7 \\ 10 \\ -4 \end{pmatrix}.$$

This vector is not in U or V , as shown by the RREF of their spans. Hence, $U \cup V$ fails closure under addition.

(b)

Yes, $U \cup W$ is **a subspace**. Since $W \subseteq U$, the union $U \cup W$ equals U , which is already a subspace. Specifically, any linear combination of vectors in $U \cup W$ remains in U , satisfying subspace properties.

(c)

To find a basis for $U \cap V$, solve $U\mathbf{x} = V\mathbf{y}$. Row reducing the augmented matrix $[U \mid -V]$ yields:

$$\text{Basis: } \left\{ \begin{pmatrix} 3 \\ 4 \\ 3 \\ 5 \\ -2 \end{pmatrix}, \begin{pmatrix} 0 \\ -2 \\ 2 \\ 2 \\ 2 \end{pmatrix} \right\}.$$

Justification:

1. The vector $\begin{pmatrix} 3 \\ 4 \\ 3 \\ 5 \\ -2 \end{pmatrix}$ is explicitly in both U and V .

2. The vector $\begin{pmatrix} 0 \\ -2 \\ 2 \\ 2 \\ 2 \end{pmatrix}$ is derived from the homogeneous solution of $U\mathbf{x} - V\mathbf{y} = \mathbf{0}$, confirming it lies in the intersection.

Question 4

(a)

The basis for V (column space of $\mathbf{A}$) corresponds to the pivot columns (1, 2, 5) in the RREF. These columns in the original matrix $\mathbf{A}$ are:

$$\left\{ \begin{pmatrix} 0 \\ 1 \\ 4 \\ -2 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 5 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 6 \\ -2 \\ 3 \end{pmatrix} \right\}.$$

Dimension: 3.

(b)

The nullspace basis vectors are derived from the RREF free variables (columns 3 and 4). Parameters s, t yield:

$$\left\{ \begin{pmatrix} -2 \\ -2 \\ 3 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 0 \\ 3 \\ 0 \end{pmatrix} \right\}.$$

(c)

The row space basis is the non-zero rows of the RREF:

$$\left\{ \left(1 \ 0 \ \frac{2}{3} \ \frac{1}{3} \ 0 \right), \left(0 \ 1 \ \frac{2}{3} \ \frac{1}{3} \ 0 \right), \left(0 \ 0 \ 0 \ 0 \ 1 \right) \right\}.$$

(d)

The intersection $Null(\mathbf{A}) \cap Row(\mathbf{A})$ contains only the zero vector. A non-zero vector in both spaces would satisfy $\mathbf{v} \cdot \mathbf{v} = 0$, which is impossible. Only the zero vector $\mathbf{0}$ is in both null space and row space.